

Project Title: B-Line Public Transit: A Data-Driven Analysis for Service Optimization

Introduction:

Public transportation plays a critical role in urban mobility, and optimizing its efficiency is essential for both economic and social development. The B-Line public transport system, serving Chico and surrounding communities, is central to the daily movement of students, workers, and residents. This project is of particular interest as it combines public service improvement with advanced data science methods. By analyzing B-Line's transit data, the study aims to uncover actionable insights to improve service quality, accessibility, and cost efficiency.

Statement of Problem:

This project addresses the need for more efficient and data-informed public transportation planning within the B-Line transit system. The specific problem is the presence of underutilized routes, which was confirmed during a meeting with B-Line staff, where they acknowledged that some routes experience consistently low ridership., rigid fixed schedules that may not align with real demand patterns, and a lack of detailed analysis on how fare and external factors affect ridership. Understanding these issues is important for enhancing service delivery and improving rider satisfaction. The research will answer questions about demand fluctuations, schedule efficiency, route utilization, and fare impacts. The objectives include developing predictive models to forecast rider demand across different times and routes using time-series methods such as AutoRegressive Integrated Moving Average (ARIMA) and Long Short-Term Memory(LSTM), as well as explanatory models like regression analysis to understand the relationship between socioeconomic factors and ridership. Additionally, Price Elasticity of Demand models will be used to predict how fare changes influence rider behavior. The project also aims to detect inefficiencies in current scheduling and route planning by analyzing key performance indicators such as on-time performance, trip duration variability, and passenger load data. These metrics, derived from historical transit data provided by B-Line, will support data-driven policy recommendations. The scope focuses on generating system-level operational insights.

Project Description:

1. Background:

The B-Line is Chico's primary bus system, serving a broad population including students at California State University (CSU) Chico and residents. There is a growing need to align transit services with community growth and usage patterns, as current service levels do not always reflect rider needs—for example, based on personal observation and discussions with B-Line staff, there are very few buses operating after 5 PM on Fridays, and some areas experience low bus frequency throughout the day. This project proposes an innovative, data-driven approach that involves analyzing historical ridership data using time series models (such as ARIMA and LSTM) to forecast demand, regression and elasticity models to evaluate fare impacts, and anomaly detection techniques to

identify underutilized routes. The results of these analyses will be integrated into an interactive , providing a visual and accessible tool for B-Line staff to support data-driven decision-making and optimize transit operations.

2. Research Aims and Objectives:

- To analyze temporal ridership patterns and predict demand fluctuations using time series forecasting models such as ARIMA and LSTM.
- To identify inefficiencies in bus schedules by comparing actual ridership during off-peak hours with schedule frequency, using exploratory data analysis (EDA) and clustering techniques.
- To detect underutilized routes and stops by applying anomaly detection methods and spatial analysis techniques to stop-level ridership data.
- To evaluate how fare changes influence ridership behavior using regression models and price elasticity of demand calculations.

3. Research Questions:

1. How does rider demand fluctuate across different times of the day, days of the week, and seasons?
2. Are there inefficiencies (low ridership relative to scheduled service) in B-Line's current fixed schedule, especially during off-peak hours?
3. Which routes and stops are underutilized, and how can this be identified using ridership data?
4. What is the impact of fare changes on ridership behavior across different user groups?

4. Literature Review:

Public transportation systems are crucial for sustainable urban mobility, yet challenges persist in optimizing transit operations and reversing ridership declines. Previous research has advanced transit optimization models, including demand-responsive scheduling (Wagale et al., 2013), heuristic optimization approaches such as Particle Swarm Optimization and Genetic Algorithms (Basak et al., 2019), and multi-period analytic models that account for demand elasticity and time-varying system conditions (Chang & Schonfeld, 1991). Additionally, studies like Polzin and Godfrey (2019) have examined the impacts of changing travel behaviors, identifying factors such as increased auto ownership . However, despite these contributions, gaps remain in developing integrated, real-time, multimodal frameworks capable of dynamically optimizing public transit systems in response to evolving urban mobility patterns. This literature review outlines prior findings, highlights converging and diverging approaches, and identifies the knowledge gap this project seeks to address.

- **Prior Research on Transit Scheduling and Ridership:**

A substantial body of research has focused on transit scheduling optimization. Wagale et al. (2013) introduced a Demand-and-Travel time Responsive (DTR) model, incorporating real-time passenger demand data to optimize bus scheduling. This approach was shown to reduce both waiting times and operational inefficiencies in dense urban environments. In a complementary study, Basak et al. (2019) applied heuristic optimization algorithms — Genetic Algorithms (GA) and Particle Swarm Optimization (PSO) to optimize on-time bus performance, concluding that PSO provided superior scheduling outcomes.

Further advancing scheduling theory, Chang and Schonfeld (1991) developed multi-period optimization models that integrate time-dependent demand fluctuations and operating costs. Their analytic models optimize system attributes such as headways, route spacing, and fares under varying demand and cost conditions, including peak and off-peak periods. Unlike earlier studies, this research considers both predictable (fixed) and changing (elastic) rider demand. It shows that keeping a consistent balance between how far apart routes are and how often buses run generally works well — even when demand goes up and down throughout the day. Their model helps transit planners adjust schedules based on daily patterns in ridership, making it a useful tool for creating more flexible and responsive service plans.

In parallel, research into ridership behavior has illuminated external factors affecting transit use. Polzin and Godfrey (2019) found that declining transit ridership in Florida was primarily driven by increased auto ownership, the rise of ride-hailing services, and seasonal and economic variations. While service improvements have been made, ridership continues to favor personal and on-demand modes, emphasizing the need for more adaptive and competitive public transit systems.

- **Similarities, Differences, and Remaining Gaps:**

A common theme across these studies is the emphasis on data-driven optimization to improve scheduling efficiency. Wagale et al. (2013) and Basak et al. (2019) focus on integrating real-time or heuristic techniques.

However, these approaches differ in scope and assumptions:

Although prior studies have advanced transit optimization through different approaches, they vary in scope and assumptions. In contrast, Polzin and Godfrey (2019) focused on behavioral factors behind ridership decline without offering operational models or predictive tools. Collectively, these studies highlight the absence of a unified, machine learning-based, real-time optimization framework that integrates multimodal options, dynamic demand data, and predictive scheduling capabilities responsive to evolving system performance and commuter

preferences. This gap justifies the need for a system that not only analyzes rider behavior but also supports operational decision-making through real-time, data-driven models — which is the aim of this project.

- **Knowledge Gaps and Project Contribution:**

Despite advances in optimization theory, heuristic modeling, and ridership analysis, there is a gap in developing a real-time, multimodal, machine learning-based optimization framework that:

Dynamically integrates real-time passenger demand, multimodal mobility options, and adaptive scheduling strategies.

Uses predictive analytics to anticipate and respond to shifts in demand patterns and service conditions.

Bridges the operational and behavioral dimensions of ridership decline through real-time, data-driven decision making.

This project aims to fill this gap by designing a machine learning-based, multimodal transit scheduling framework, where multimodal refers to integrating different types of transportation options — such as fixed-route buses, on-demand shuttles, and potential micromobility services — into scheduling decisions.

5. Project Plan The project will follow these steps:

- Obtain necessary datasets from B-Line
- Clean and preprocess the data
- Apply time series and clustering models to analyze demand
- Use regression and elasticity models to assess fare impacts
- Summarize findings and develop recommendations for B-Line

6. Deliverables:

The project will include interactive dashboards and static visualizations to effectively communicate key insights. These visual tools will present:

- **Ridership trends over time** – including daily, weekly, and seasonal patterns
- **Route and stop-level utilization** – highlighting the most and least used segments of the system
- **Fare impact summaries and demand forecasts** – illustrating how fare changes influence ridership and projecting future demand patterns

These visuals will be designed to support data-driven decision-making for B-Line administrators and to make the results easily interpretable

Explanation of Data:

The following data is needed:

- Historical ridership logs (which give ridership data by location.
- Bus route and schedule information
- Fare structure changes over time
- Operational records (delays, breakdowns, etc.)

I have now received the data from the B-Line team and have completed the initial review and organization of the datasets.

Timeline for your project:

Month	Tasks
March 2025	- Requested data from B-Line on March 3rd - Met with B-Line on March 17th to discuss the project and data needs
April 2025	- Receive data from B-Line (week of April 14th) - Begin reviewing, organizing, and understanding the dataset
May 2025	- Continue exploring and familiarizing with the data - Start data cleaning and preprocessing
June 2025	- Conduct Exploratory Data Analysis (EDA) - Identify key variables and potential data gaps
July 2025	- Develop and apply predictive models (e.g., ARIMA, LSTM) - Begin route optimization analysis
August 2025	- Build regression and elasticity models - Analyze fare impacts and schedule inefficiencies
September 2025	- Create visualizations for ridership trends, route utilization, and heatmaps
October 2025	- Refine all visual outputs
November 2025	- Draft final report and compile findings - Begin creating presentation materials
December 2025	- Submit draft to advisor for feedback - Revise report and visuals based on feedback

- January 2026**
 - Finalize report and dashboards
 - Schedule and prepare for final presentations
- February 2026**
 - Deliver presentation to B-Line team and academic committee
 - Make final adjustments to deliverables
- March–April 2026**
 - Submit final report and materials
 - Complete any final academic or administrative requirements

Literature Cited:

Bibliography

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- WAGALE, M., SINGH, A. P., SARKAR, A. K., & ARKATKAR, S. (2013). Real-Time Optimal Bus Scheduling for a City using A DTR. doi:<https://doi.org/10.1016/j.sbspro.2013.11.179>

Graduate Committee Approval: Please have each of your committee members read and sign this document.

I have read _____'s project proposal and find it adequate and I agree to serve on their graduate advisory committee.

Committee Chair:

Robin Donatello

Printed

Signed

Date

Second Member:

Kathy Gray

Printed

Signed

Date

Third Member (if applicable):

Printed
Date

Signed